# QUINN AHO

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## EDUCATION

**Wayne State University** **Detroit, MI**

*B.S. in Computer Science Engineering:* *Aug 2022 – May 2026*

- GPA: 3.86
- Coursework: Computer Graphics, Graphical + Geometric Modeling, Game Programming, Intro to LLMs, Software Engineering.

Honors: **Congressional App Challenge Winner**, XR Midwest / HackDearborn Finalist, Edward + Bernard Gaffney Scholar.

## WORK EXPERIENCE

**Johns Hopkins Applied Physics Lab** **Laurel, MD**

*SWE Intern - Immersive Technologies and Software Solutions (Extended Engagement):* *May 2024 – Present*

- Deployed real-world military training simulations in **Unity** for active defense programs.
- **Independently prototyped next-generation immersive trainers in Unreal Engine**, one prototype **received internal honors and secured five years of continued program funding.**
- Built a node-based authoring system enabling non-programmers to build structured mission logic directly in-editor.
- Optimized real-time rendering performance through shader batching, lightmap strategies, and custom editor-side profiling workflows.

**Freelance Contract - NextGenCoding** **Remote, USA**

*Unreal Engineer - Case Study :* *Sep 2024 – Nov 2024*

- Integrated Nvidia Audio2Face and ElevenLabs into Unreal Engine to drive **real-time AI avatar animation** with synchronized lip-sync, speech, and procedural motion.
- Built optimized animation state machines powering interactive MetaHuman experiences, shipped for a high-profile client demo.

**Corvid Technologies** **Mooresville, NC**

*VR Visualization Intern:* *May 2023– Aug 2023*

- Developed **Unreal Engine VR visualization tools** and **IK-based avatar systems** using C++ and Blueprints.
- **Solo-produced an interactive cinematic demo** integrating **CAD-sourced geometry, presented directly to a defense stakeholder**.
- Improved debugging and profiling pipelines for real-time VR simulation environments.

**GS Engineering** **Houghton, MI**

*Simulation Engineering Intern:* *Nov 2023– May 2024*

- Built a custom **Unreal Engine plugin for KPI visualization in vehicle digital twin simulations**.
- Focused on system design for collisions, physics tuning, and gameplay logic for realistic scenarios.
- Created editor-friendly tools to support engineering workflows.

## PROJECTS

**Glint 3D – C++ Automated 3D Rendering Engine**

*Portfolio Project - GitHub* *Feb 2025 – Present*

- **Built a C++ 3D engine from scratch,** interactive UI, scriptable CLI, JSON ops pipeline, and headless batch rendering.
- Engineered a **multi-format asset pipeline (glTF, FBX, OBJ, PLY)** with PBR shading and BVH-accelerated raytracing.
- **Dual-use design: human-driven** scene editing through the UI, **automation-driven** workflows through CLI and headless rendering.

**Blueprint Health Analyzer - Unreal Engine's First-Ever Blueprint Linter**

*Side Hustle - Documentation Demo Video* *Nov 2025 – Present*

- **Created the first dedicated Blueprint linting plugin for Unreal Engine**, identifying quality issues and enforcing best practices.
- Led a successful **20+ person beta program,** iterating on feedback to ship production-quality analysis rules and polished UX.
- Built full web documentation with interactive Blueprint examples and video walkthroughs.

**AI Niagara Plugin – Natural Language VFX Control** *Jan 2025 – Present*

*Portfolio Project - Under Development*

- Building an **Unreal Engine plugin that generates production-ready particle VFX from natural language.**
- Full editor integration: Slate UI, cloud + local AI backends, and a custom translation pipeline from LLM output to live Niagara systems.

## SKILLS

**Languages**: C++, C#, Python, GLSL/HLSL, JSON, Blueprint

***Engines & Frameworks:*** *Unreal Engine 4/5, Unity, Custom Engine (OpenGL, Vulkan, WebGPU)*

***3D & Graphics:*** *OpenGL, DirectX, Vulkan, WebGL2, WebGPU, PBR Rendering, Raytracing, Shader Development*

***Pipelines & Data:*** *CAD Integration, Blender, SolidWorks, NX, CI/CD, Git, Perforce*

***Specialties:*** *3D Visualization Platforms, VR/AR Systems, Digital Twin Simulation, Custom Tooling & Plugins, Technical Art*